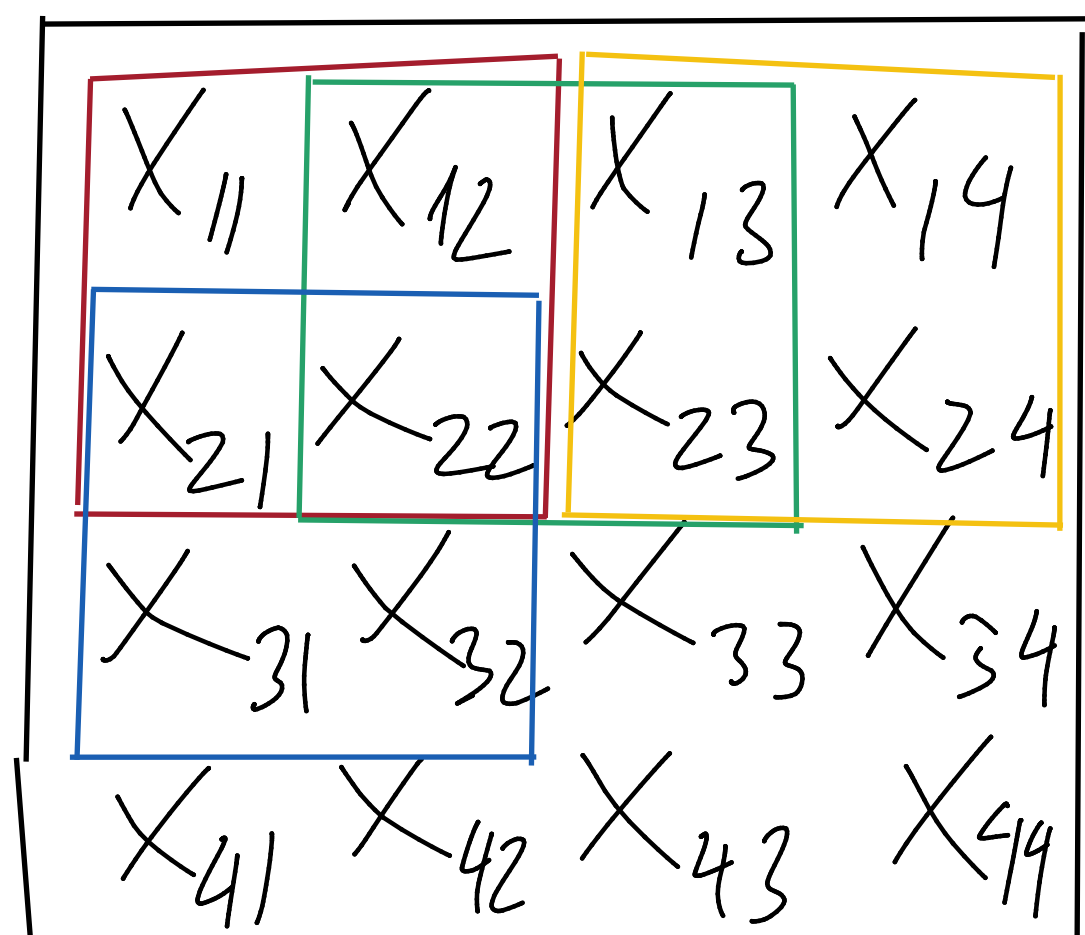


Lecture 4

Inductive bias

Convolutional Neural Networks

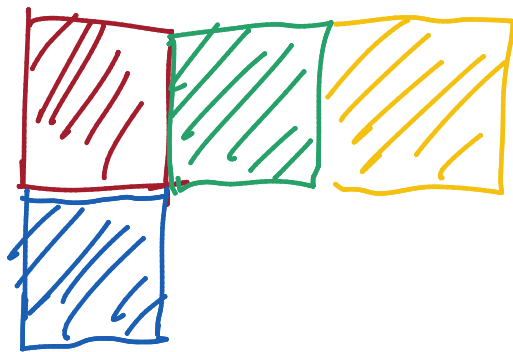
$H \times W \times 3$



$\rightarrow 0 - 255 \Rightarrow [0, 1]$

$\begin{bmatrix} \alpha_{11} & \alpha_{12} \\ \alpha_{21} & \alpha_{22} \end{bmatrix} \leftarrow \text{kernel filter}$

$$g = \sum_{i,j} \alpha_{ij} x_{ij}$$



Convolution

$$\frac{1}{9} \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

box blur

$$\frac{1}{8} \begin{bmatrix} -1 & -1 & -1 \\ -1 & 8 & -1 \\ -1 & -1 & -1 \end{bmatrix}$$

$$\frac{(-8 + 8)}{8} = 0$$

$$\begin{matrix} \alpha_{11} & \alpha_{12} & \alpha_{13} \\ \alpha_{21} & \dots & \dots \\ \alpha_{31} & & \end{matrix}$$

α - learnable parameters

$$\begin{matrix} X_{11} & X_{12} & X_{13} & X_{14} \\ X_{21} & X_{22} & X_{23} & X_{24} \\ X_{31} & X_{32} & X_{33} & X_{34} \end{matrix}$$

*

$$\begin{matrix} \alpha_{11} & \alpha_{12} \\ \alpha_{21} & \alpha_{22} \end{matrix}$$

$$\begin{pmatrix} \alpha_{11} & \alpha_{12} & 0 & 0 & \alpha_{21} & \alpha_{22} & 0 & \dots & 0 \\ 0 & \alpha_{11} & \alpha_{12} & 0 & 0 & \alpha_{21} & \alpha_{22} & 0 & \dots \\ 0 & 0 & \alpha_{11} & \alpha_{12} & 0 & 0 & \dots & & \\ 0 & 0 & 0 & 0 & \alpha_{11} & \alpha_{12} & & & \\ 0 & 0 & 0 & 0 & 0 & 0 & \alpha_{11} & & \\ \vdots & & & & & & & & \\ 0 & \dots & 0 & \alpha_{11} & \alpha_{12} & 0 & 0 & \alpha_{21} & \alpha_{22} \end{pmatrix} \begin{pmatrix} X_{11} \\ X_{12} \\ X_{13} \\ X_{14} \\ \vdots \\ X_{31} \\ X_{32} \\ X_{33} \\ X_{34} \end{pmatrix} *$$

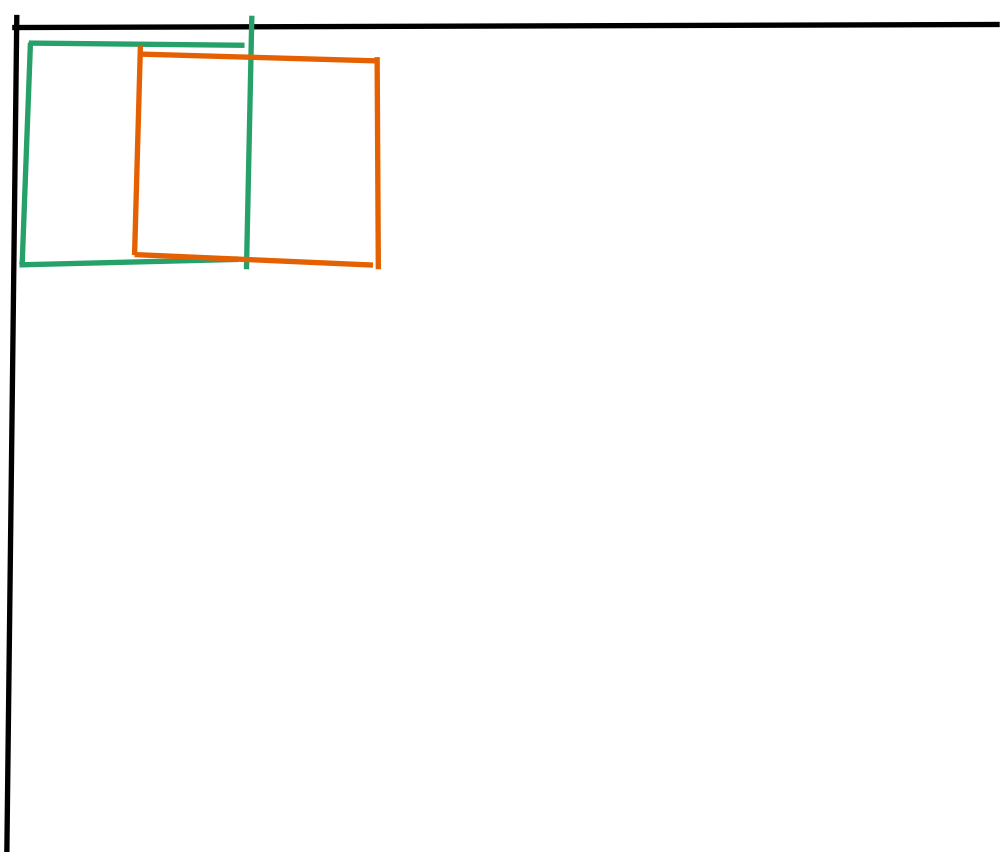
- kernel size (k)

2×2 , 3×3 , $k_1 \times k_2$

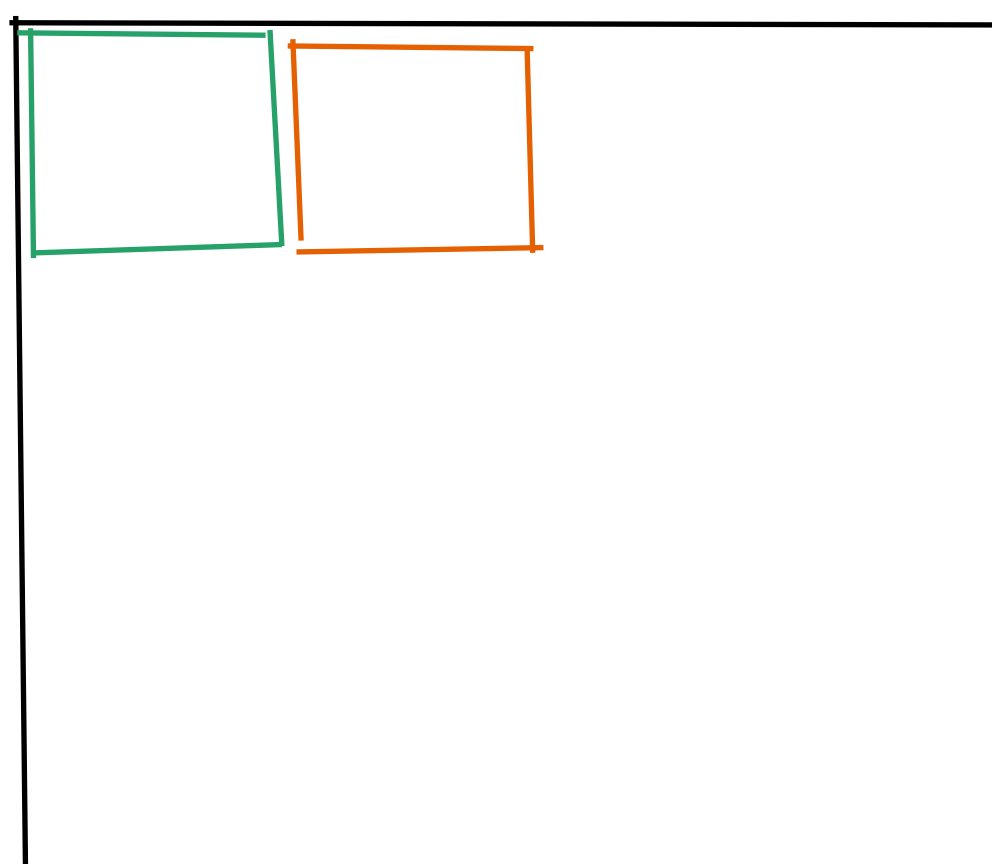
$k_1 k_2$ -
num params

- Stride (s)

$s=1$



$s=2$

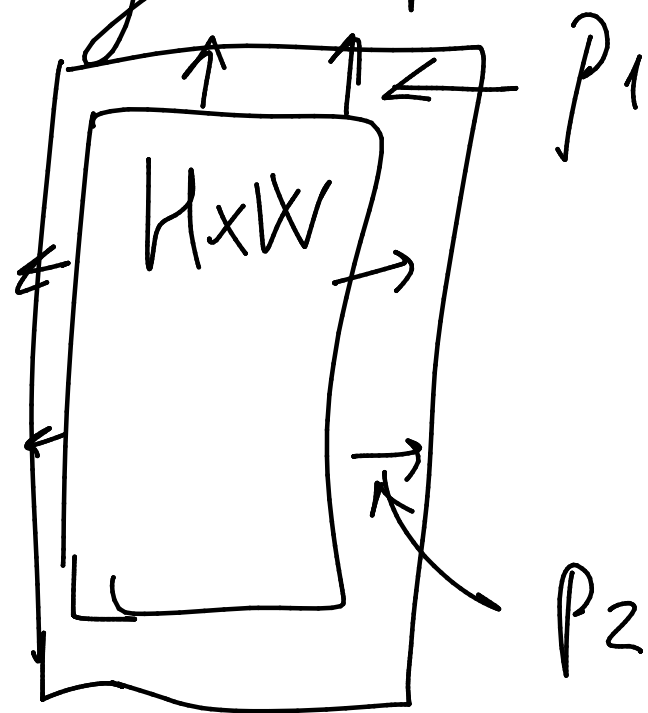


$H \times W$, $k_1 \times k_2$, s_1 , s_2

$$H_{out} = \left\lfloor \frac{H - k_1 + 1}{s_1} \right\rfloor$$

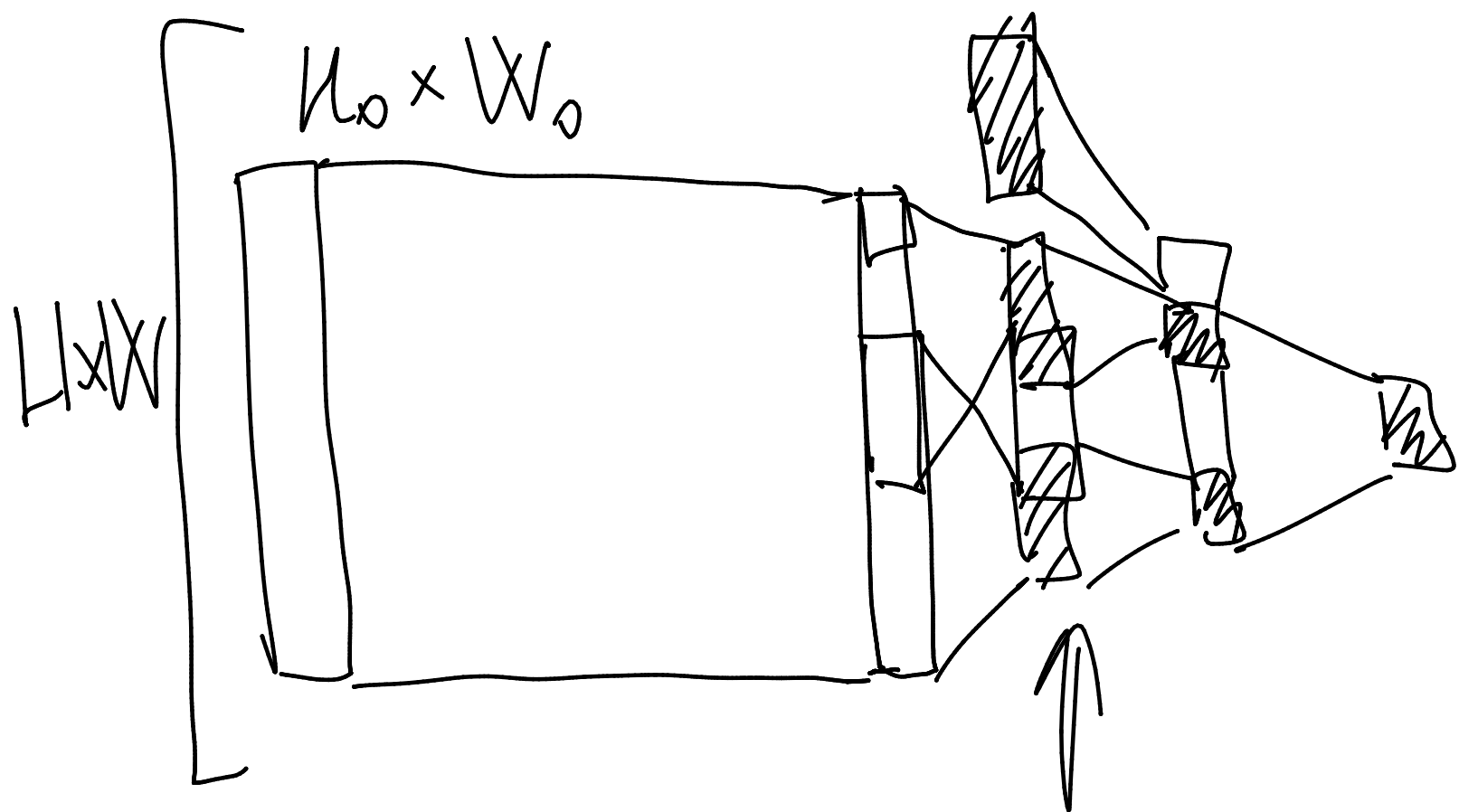
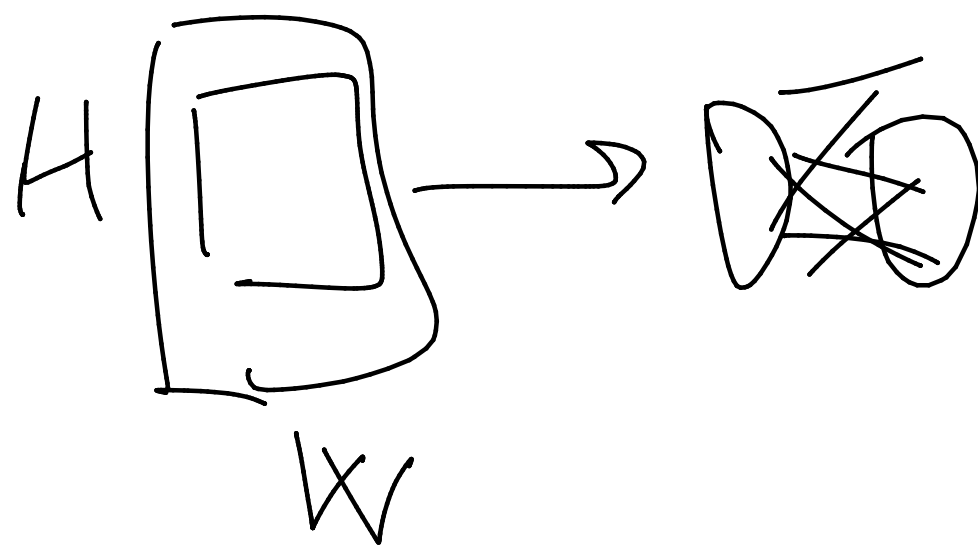
$$W_{out} = \left\lfloor \frac{W - k_2 + 1}{s_2} \right\rfloor$$

- padding (p)



$$p = \left\lfloor \frac{k - 1}{2} \right\rfloor \quad s=1$$

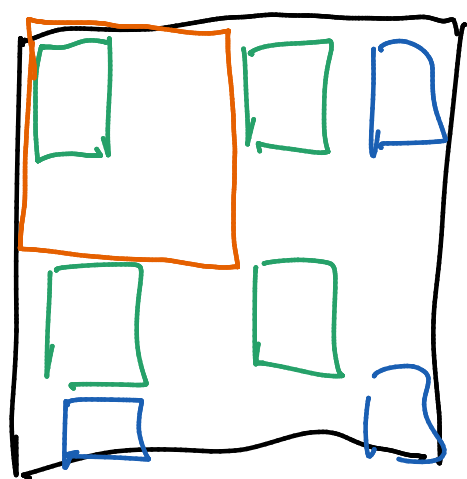
Receptive Field



$$H_0 \leq H$$

$$W_0 \leq W$$

• Dilation (d)

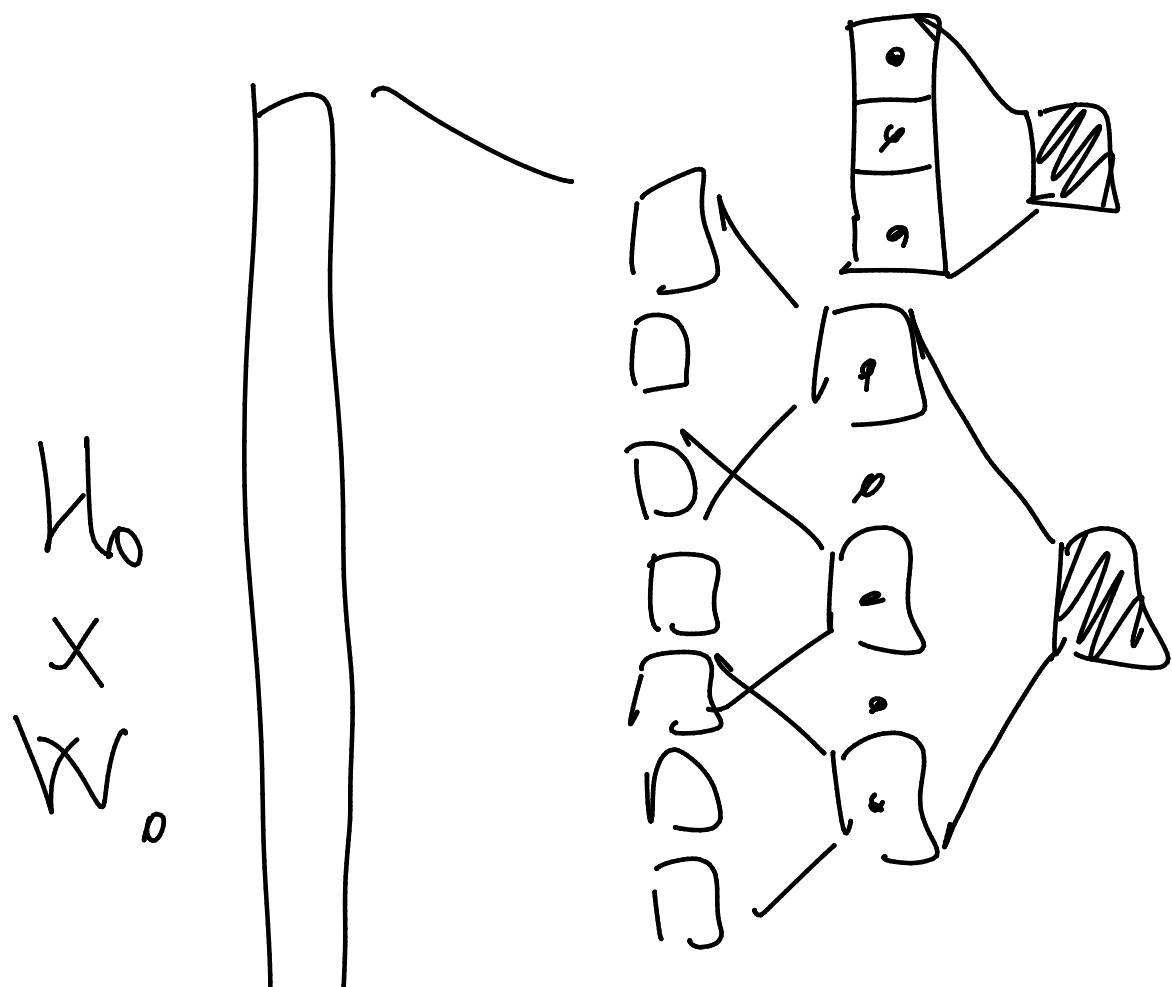
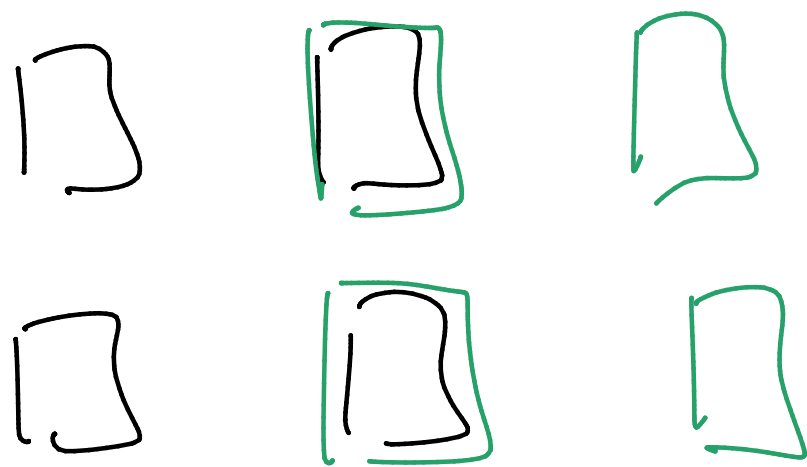
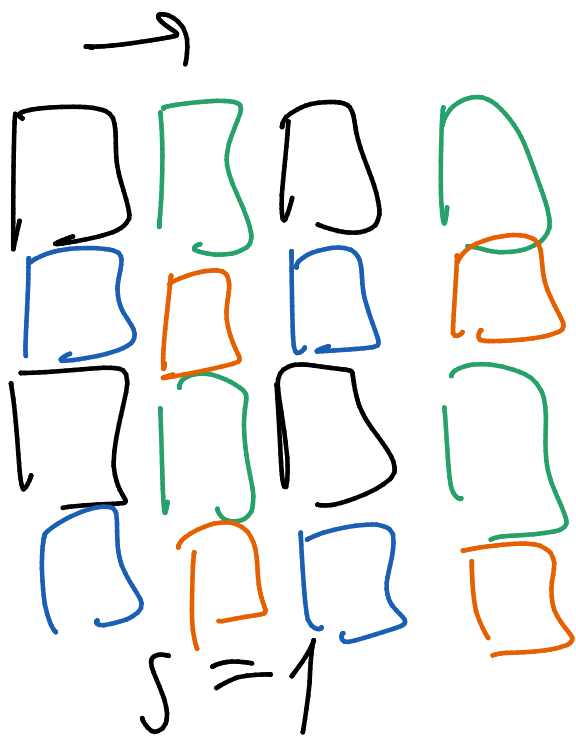


$$d=1$$

$$d=2$$

$$d=3$$

$$d_{11}x_{11} + d_{12}x_{13} + d_{21}x_{31} + d_{22}x_{33}$$



normal

feature map

$$X: C_{in} \times H_{in} \times W_{in}$$

input

$$y: C_{out} \times H_{out} \times W_{out}$$

output

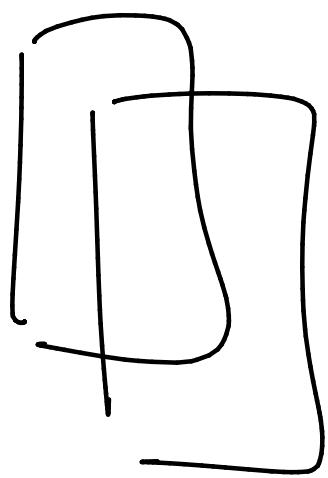
$$W: C_{in} \times C_{out} \times K_H \times K_W$$

$$1 \leq i \leq C_{out}$$

$$1 \leq j \leq C_{in}$$

$$y^i = \sum_{j=1}^{C_{in}} x_j * w_{ji}$$

$$X: 2 \times 2 \times 2$$



$$y: 3 \times 2 \times 2$$

$$X_1 = \begin{bmatrix} 2 \times 2 \end{bmatrix} \quad X_2 = \begin{bmatrix} 2 \times 2 \end{bmatrix}$$

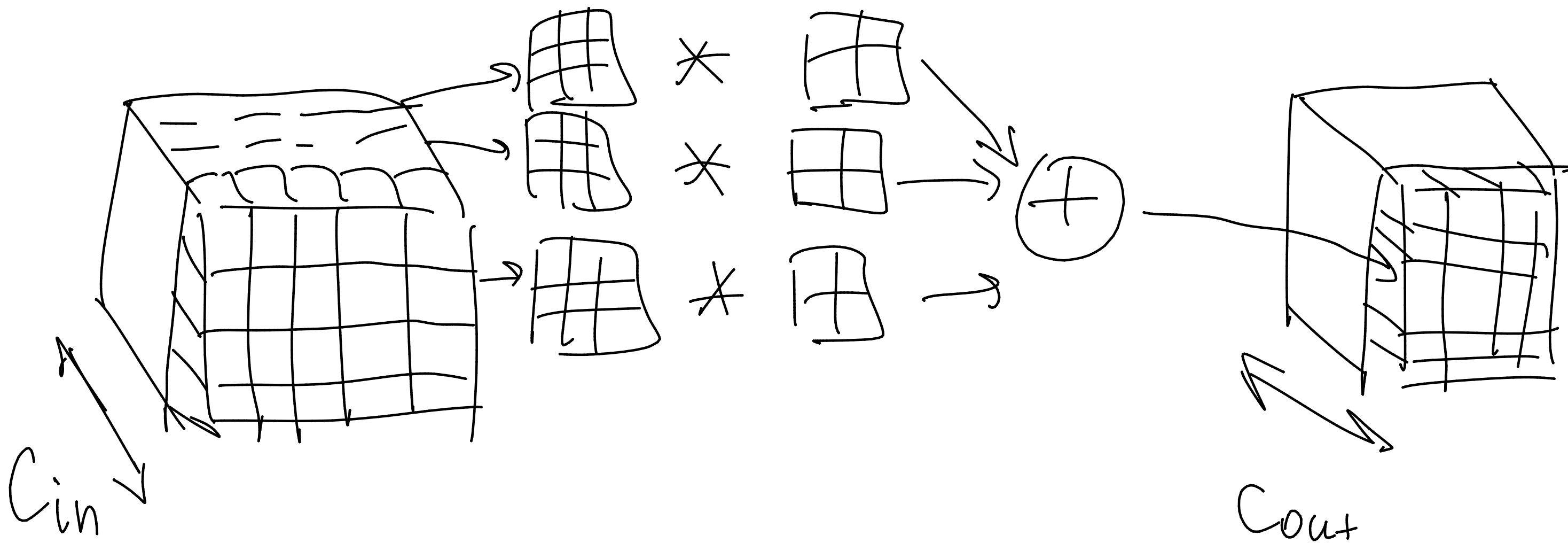
$$K_H = K_W = 1$$

$$\alpha_{ji}: 1 \times 1$$

$$\alpha: 2 \times 3 \times 1 \times 1$$

$$1 \leq i \leq 2$$

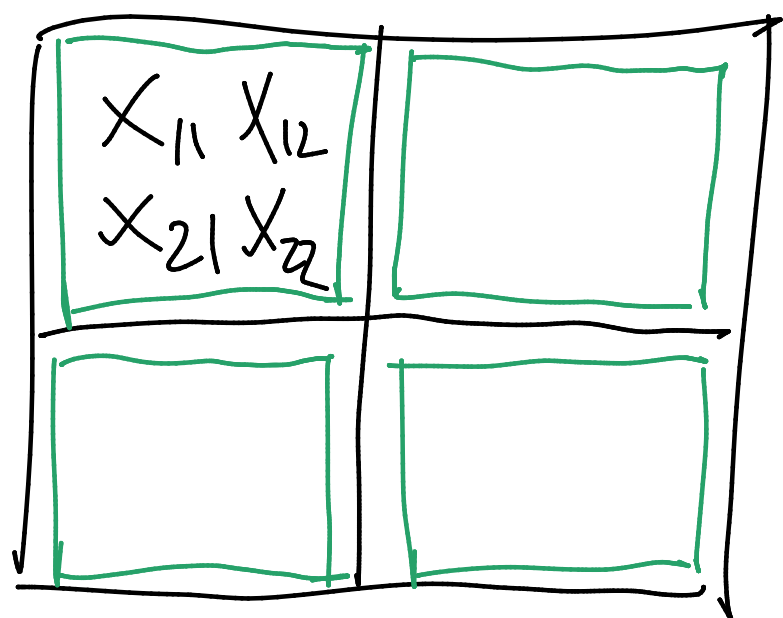
$$y_i = \alpha_{1i} \cdot X_1 + \alpha_{2i} \cdot X_2 \quad 1 \leq j \leq 3$$



$$X_j * w_{ji} + b_i \quad b \in \mathbb{R}^{C_{out}}$$

• Pooling

($k=s$)



• Max Pooling

$$y = \max_{i,j} x_{ij}$$

$$\max \{ x_{11} \dots x_{22} \} \rightarrow \text{shaded box}$$

• Average Pooling

$$y = \frac{1}{k^2} \sum_{i,j} x_{ij}$$

$$\text{avg} \{ x_{11} \dots x_{22} \}$$

